

Sheng-Kai (Kyle) Wen

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Education

University of Massachusetts Amherst

M.S. in Computer Science. *Coursework: Distributed Systems, ML, NLP, Information Retrieval*

Sep 2024 – May 2026

National Tsing Hua University

M.S. in Physics. *Coursework: Operating Systems, Computer Architecture, Databases*

Sep 2020 – Jun 2023

National Dong Hwa University

B.S. in Electrical Engineering. *Coursework: Algorithms, Data Structures*

Sep 2016 – Jun 2020

Experience

Honda Research Institute (HRI) & 99P Labs

Graduate Research Extern (Advised by Prof. Andrew McCallum)

Feb 2026 – May 2026

- **Improved repo-level code retrieval** on DeepCodeBench by **architecting a Graph-RAG framework** that fuses Knowledge Graphs with dense retrievers to model function call graphs and structural dependencies.
- **Enabled precise structural localization** across large codebases by **building an AST-based indexing pipeline**, closing recall gaps in standard BM25 and CodeRankEmbed retrievers.
- **Enriched LLM reasoning context** for code repair by **engineering a LocAgent-based test-linkage layer** that traces failing tests back to specific source locations, grounding model inference in verifiable evidence.
- **Accelerated ablation throughput** by **designing a modular QA evaluation harness** integrating Graph-RAG prototypes with large-scale codebases, enabling systematic comparison of retrieval strategies.

PuckAI — NHL Analytics Platform

Machine Learning Engineer — UMass CICS (CICStep)

May 2025 – Present

- **Delivered live analysis for 6K+ NHL skaters spanning five decades (1968–present)** by **engineering a full-stack platform** (Next.js, FastAPI, PostgreSQL + pgvector, Supabase) with AI scouting reports via Claude Opus 4.8 and Voyage AI RAG.
- **Tiered skaters Elite→Fringe across eras** by **building a position-aware, era-adjusted percentile system** (per-season z-scores), and projected young provisional players via per-position **XGBoost** on early-career windows.
- **Served sub-millisecond autocomplete** for 6K+ players by **implementing an in-memory PrefixTrie** rebuilt on startup, eliminating per-keystroke DB hits entirely.
- **Ensured daily data freshness** for 6K+ profiles by **building a 7-stage modular ETL pipeline** from NHL.com and EliteProspects, with idempotent ON-CONFLICT upsert for in-season refresh.

Fan2see, Taipei, Taiwan

Co-Founder & Full Stack Engineer

Apr 2021 – Apr 2022

- **Scaled to hundreds of active users** by **building a full-stack video platform** (React, Node.js) with a modular architecture supporting concurrent streaming and real-time content management.
- **Reduced media processing latency by 25%** by **architecting an asynchronous AWS Lambda/S3 pipeline** for high-volume video uploads and multi-stage data transformations.
- **Achieved 30% month-over-month user growth** by **refactoring a monolithic backend into a PERN stack** and optimizing PostgreSQL schemas, improving scalability by 40%.

Projects

MAIN-RAG: Multi-Agent Filtering in Retrieval-Augmented Generation

Sep 2025 – Dec 2025

- **Cut LLM context size by 40–60%** without sacrificing recall on HotpotQA and FEVER by **designing a multi-agent reranking pipeline** that filters retrieval noise before generation.
- **Raised retrieval precision and QA accuracy** by **implementing logit-margin Cross-encoders** scoring Doc–Query–Answer triplets across multi-hop benchmarks.
- **Raised nDCG@10 on multi-hop QA** by **integrating Hybrid Search** (BM25 + Neural) with cross-encoder re-ranking, reducing hallucination in noisy retrieval environments.

Hierarchical Video Summarization with Temporal Alignment

Sep 2025 – Dec 2025

- **Raised ROUGE-L by 80%** (35.9% → 64.1%) by **fine-tuning BART** on MeetingBank and applying sliding-window chunking in LongT5 for long-document coherence.
- **Achieved 80% temporal alignment accuracy** by **training a BERT-based module** to map summary segments to video timestamps, ensuring factual consistency in multimodal content.

Technical Skills

AI & Information Retrieval: RAG, Hybrid Search, Knowledge Graphs, PyTorch, XGBoost, pgvector

Web & Systems: Next.js, FastAPI, React, Node.js, PostgreSQL, Docker, AWS (Lambda/S3), Supabase

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL